## SUMMARY:

Stocks, Equity ⇒ Yahoo finance

Dataframe Objects in Pandas.

| Date | AAPL | MCFT | TSLA | GE |
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LA: $A \in \mathbb{R}^{m \times n}$

$A: \mathbb{R}^n \longmapsto \mathbb{R}^m$

$A\vec{x} = \vec{b}$

$A \in \mathbb{R}^{2\times 2}$

$$\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

linear transformation.

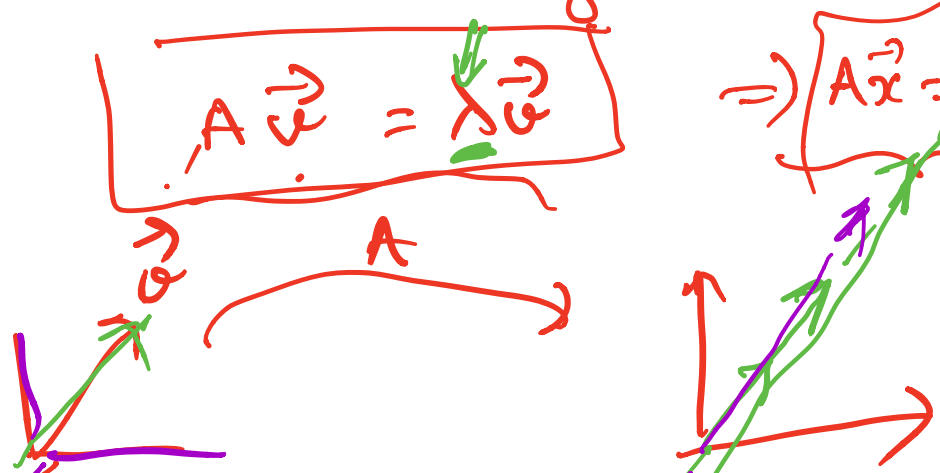
# Eigenvalues / Eigenvectors

For a matrix $A \in \mathbb{R}^{n \times n}$

$\lambda \in \mathbb{R}$, vector $\vec{v}$

Eigvalue  Eigenvector

$$\boxed{A\vec{v} = \lambda\vec{v}} \Rightarrow \boxed{A\vec{x} = b}$$




A takes a vector in $\mathbb{R}^3$ & its transforms it to another vector $\vec{w}$

$\Rightarrow (\lambda, \vec{v})$ always exist in pairs.

Existence:

Do all matrices have ~~EVs~~ $\lambda$'s & $\vec{v}$'s

$$A\vec{v} = \boxed{\lambda \vec{v}} \Rightarrow \underline{(A - \lambda I)\vec{v}} = 0$$

characteristic poly, $\vec{v}$'s

There is always going to sol$^n$.

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Summary:

Existence is clear.

$\lambda$ & $\vec{v}$ have the same geometrical meaning where the linear transformation is a scaling!

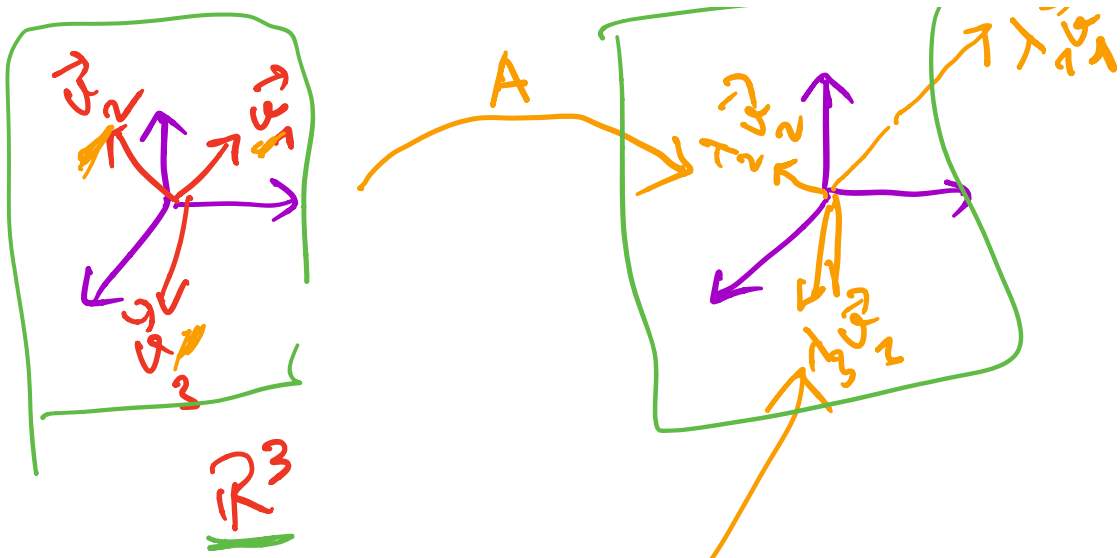
Def$^n$: $A \in \mathbb{R}^{m \times m}$ is diagonalizable

$$\boxed{A} = U \boxed{D} U^{-1}$$

orthonormal ($UU^T = I$)

diagonal

$$\begin{bmatrix} x & x & x & x \\ x & x & x & x \\ x & x & x & x \\ x & x & x & x \end{bmatrix} \qquad \begin{bmatrix} x & & 0 & \\ & x & & \\ 0 & & x & \\ & & & x \end{bmatrix}$$




$$U = \begin{bmatrix} \vec{v}_1 & \vec{v}_2 & \vec{v}_3 \end{bmatrix}$$

$$AU = \begin{bmatrix} \lambda_1\vec{v}_1 & \lambda_2\vec{v}_2 & \lambda_3\vec{v}_3 \end{bmatrix}$$

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decomposition

$A \in \mathbb{R}^{n\times n} \Rightarrow \boxed{A = PDP^{-1}}$ ? $\begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$ $\in \mathbb{R}$

$A \in \mathbb{R}^{m\times n} \Rightarrow A = \underbrace{U}_{m\times m}\,\underbrace{\Sigma}_{m\times n}\,\underbrace{V^T}_{n\times n}$

Symmetric matrix:

**Def$^n$**: Symmetric matrix $\Rightarrow$ if $A = A^T$
$A \in \mathbb{R}^{n \times n}$ (mirror image of itself across the diagonal).

**THM**: All symmetric matrixes are (orthonormally) diagonalize

$\Downarrow$

col$^n$ of stock price

all sym. matrice have real Eigenvalue

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Singular Value Decomposition.

Let $A \in \mathbb{R}^{m \times n}$ matrix. There always $\exists$ orthonormal matrices $U, V$ such that

$$A = \underset{m \times m}{U} \; \overset{m \times n}{\Sigma} \; \overset{n \times n}{V^T}$$

diagonal matrix

can be a non-square

"singular values"

$m \times n$ $m > n$

$$\Sigma = \begin{bmatrix} \sigma_1 & & 0 \\ & \ddots & \\ 0 & & \sigma_n \\ & 0 & \end{bmatrix}$$

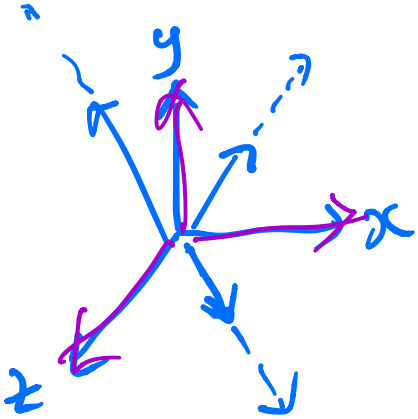
$m < n$

$$\Sigma = \begin{bmatrix} \sigma_1 & & & 0 \\ & \ddots & & \\ 0 & & \sigma_m & \end{bmatrix}$$

$m = 4$ $n = 3$

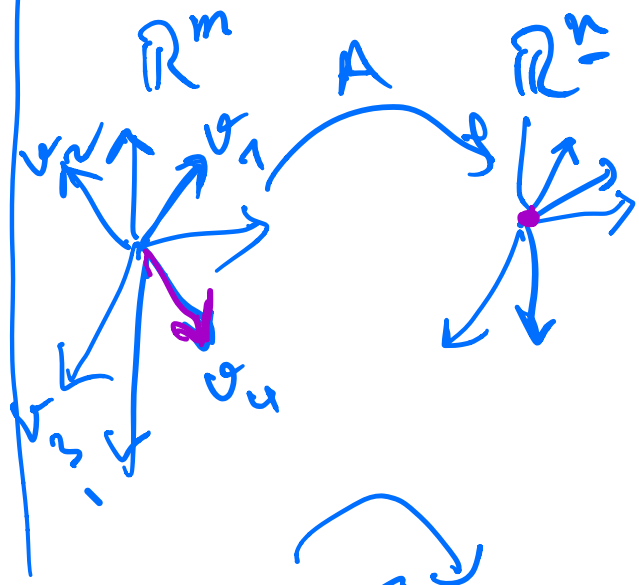
**EVD** $(A = PDP^{-1})$

$n \times n$ matrix




more powerful
more restriction

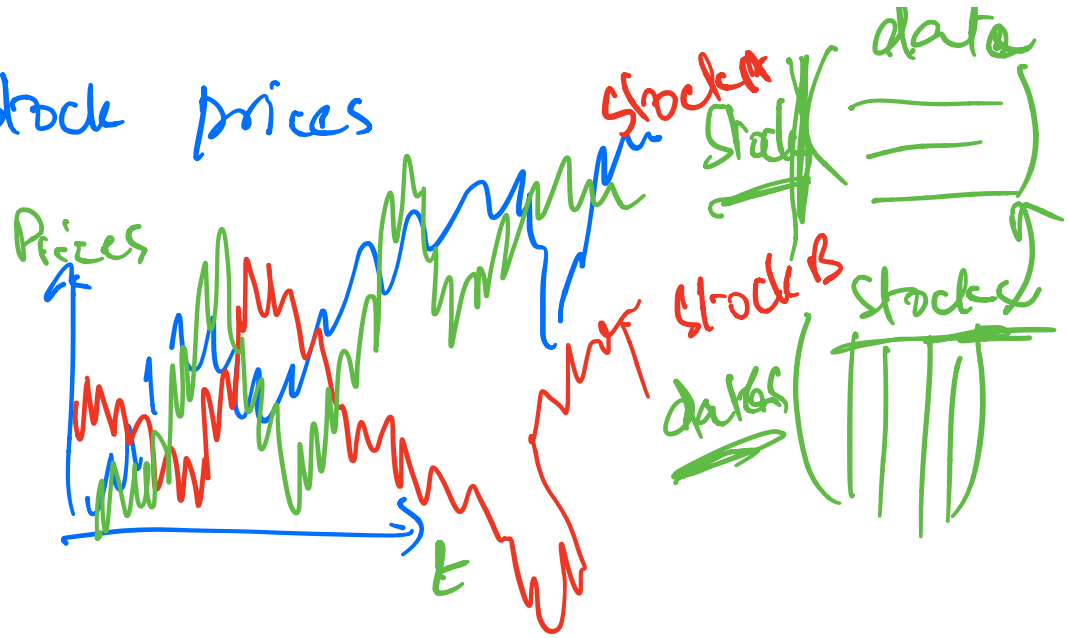
**SVD** $(A = U\Sigma V^T)$




$$AU = \Sigma V$$

general
but always
available

Stock prices




Covariance of our matrix A

$$A^TA \Longleftrightarrow AA^T$$

EVD $\Rightarrow$ of covariance $\frac{1}{n-1}(A^TA) \Rightarrow QDQ^T$

SVD $\Rightarrow$ " $A^TA = U\Sigma^2U^T$

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Example: let $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$

$$A^TA = \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$$

$$\tilde{A} = \begin{bmatrix} x & x & x \\ x & x & x \\ x & x & x \end{bmatrix}$$

Eigenvalues of $\tilde{A}$

$$\lambda = \boxed{0}, \underbrace{9, 25}$$

Eigenvectors =